

# SupplementaryMaterial

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2026-07-08

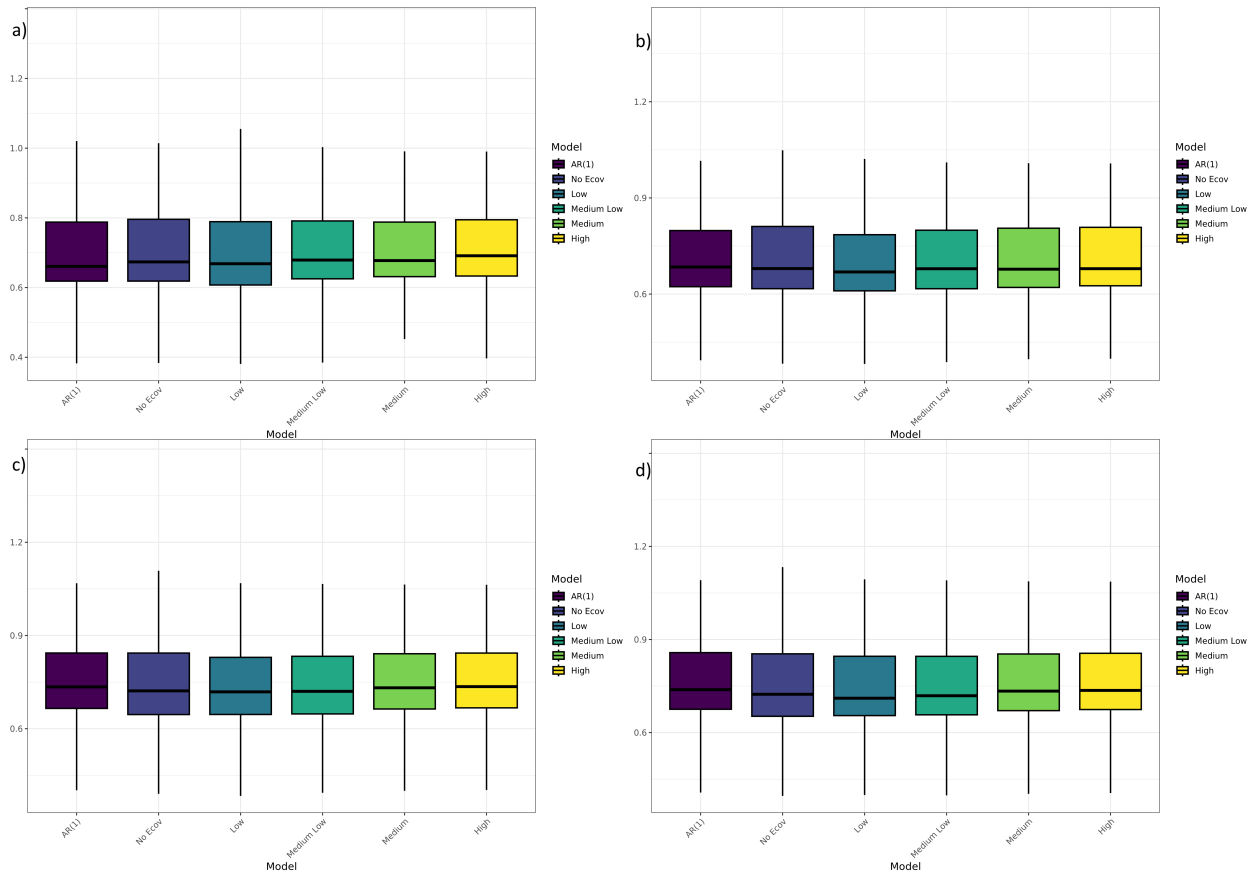


Figure 1: Annual variation in catch of each EM for a) the low-temperature OM scenario, b) the medium-low temperature OM scenario, c) the medium-temperature scenario, and d) the high-temperature OM scenario

Table 1: Convergence rates of each MSE scenario tested.

| Experiment | OM                              | EM                 | Percent Non-Converged |
|------------|---------------------------------|--------------------|-----------------------|
| Projection | Normal Projection               | AR(1)              | 0.04                  |
| Projection | Normal Projection               | No Ecov            | 0.04                  |
| Projection | Normal Projection               | Historical Average | 0.10                  |
| Projection | Normal Projection               | Terminal Year      | 0.07                  |
| Projection | Normal Projection               | Recent Window      | 0.04                  |
| Projection | Normal Projection               | Recent Trend       | 0.06                  |
| Projection | Normal Projection               | 'Good' Projection  | 0.07                  |
| Projection | Normal Projection               | 'Bad' Projection   | 0.02                  |
| Projection | NAA Reduced Random Effects      | AR(1)              | 0.04                  |
| Projection | NAA Reduced Random Effects      | No Ecov            | 0.05                  |
| Projection | NAA Reduced Random Effects      | Historical Average | 0.04                  |
| Projection | NAA Reduced Random Effects      | Terminal Year      | 0.02                  |
| Projection | NAA Reduced Random Effects      | Recent Window      | 0.03                  |
| Projection | NAA Reduced Random Effects      | Recent Trend       | 0.10                  |
| Projection | NAA Reduced Random Effects      | 'Good' Projection  | 0.02                  |
| Projection | NAA Reduced Random Effects      | 'Bad' Projection   | 0.07                  |
| Projection | 5-Year Projection               | AR(1)              | 0.00                  |
| Projection | 5-Year Projection               | No Ecov            | 0.00                  |
| Projection | 5-Year Projection               | Historical Average | 0.00                  |
| Projection | 5-Year Projection               | Terminal Year      | 0.00                  |
| Projection | 5-Year Projection               | Recent Window      | 0.00                  |
| Projection | 5-Year Projection               | Recent Trend       | 0.00                  |
| Projection | 5-Year Projection               | 'Good' Projection  | 0.00                  |
| Projection | 5-Year Projection               | 'Bad' Projection   | 0.00                  |
| Bias       | High Temperature Scenario       | AR(1)              | 0.00                  |
| Bias       | High Temperature Scenario       | No Ecov            | 0.00                  |
| Bias       | High Temperature Scenario       | High Bias          | 0.01                  |
| Bias       | High Temperature Scenario       | Medium Bias        | 0.03                  |
| Bias       | High Temperature Scenario       | Medium-Low Bias    | 0.00                  |
| Bias       | High Temperature Scenario       | Low Bias           | 0.01                  |
| Bias       | Medium Temperature Scenario     | AR(1)              | 0.06                  |
| Bias       | Medium Temperature Scenario     | No Ecov            | 0.05                  |
| Bias       | Medium Temperature Scenario     | High Bias          | 0.02                  |
| Bias       | Medium Temperature Scenario     | Medium Bias        | 0.06                  |
| Bias       | Medium Temperature Scenario     | Medium-Low Bias    | 0.06                  |
| Bias       | Medium Temperature Scenario     | Low Bias           | 0.02                  |
| Bias       | Medium-Low Temperature Scenario | AR(1)              | 0.04                  |
| Bias       | Medium-Low Temperature Scenario | No Ecov            | 0.01                  |
| Bias       | Medium-Low Temperature Scenario | High Bias          | 0.00                  |
| Bias       | Medium-Low Temperature Scenario | Medium Bias        | 0.01                  |
| Bias       | Medium-Low Temperature Scenario | Medium-Low Bias    | 0.04                  |
| Bias       | Medium-Low Temperature Scenario | Low Bias           | 0.02                  |
| Bias       | Low Temperature Scenario        | AR(1)              | 0.09                  |
| Bias       | Low Temperature Scenario        | No Ecov            | 0.14                  |
| Bias       | Low Temperature Scenario        | High Bias          | 0.08                  |
| Bias       | Low Temperature Scenario        | Medium Bias        | 0.05                  |
| Bias       | Low Temperature Scenario        | Medium-Low Bias    | 0.07                  |
| Bias       | Low Temperature Scenario        | Low Bias           | 0.03                  |

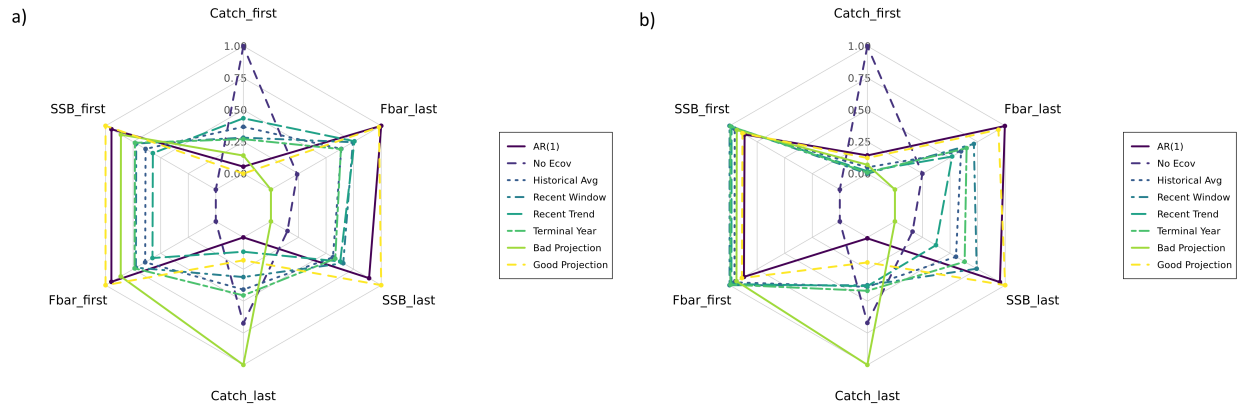


Figure 2: The performance of EM projections in the first and last 10 years of the simulation period for the a) reduced variation on Numbers at Age OM and b) the five-year assessment period

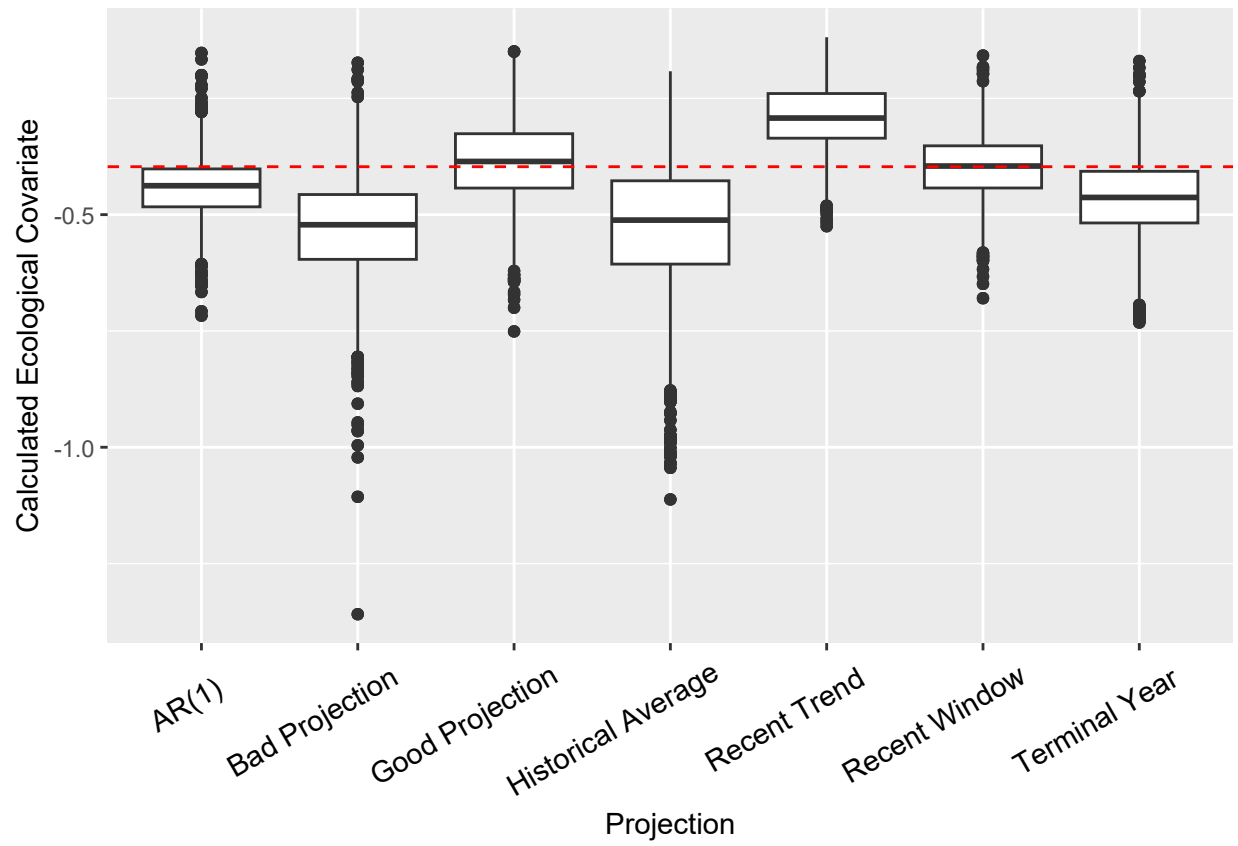


Figure 3: Estimated ecological covariates for the different temperature projections with reduced random effects on recruitment.

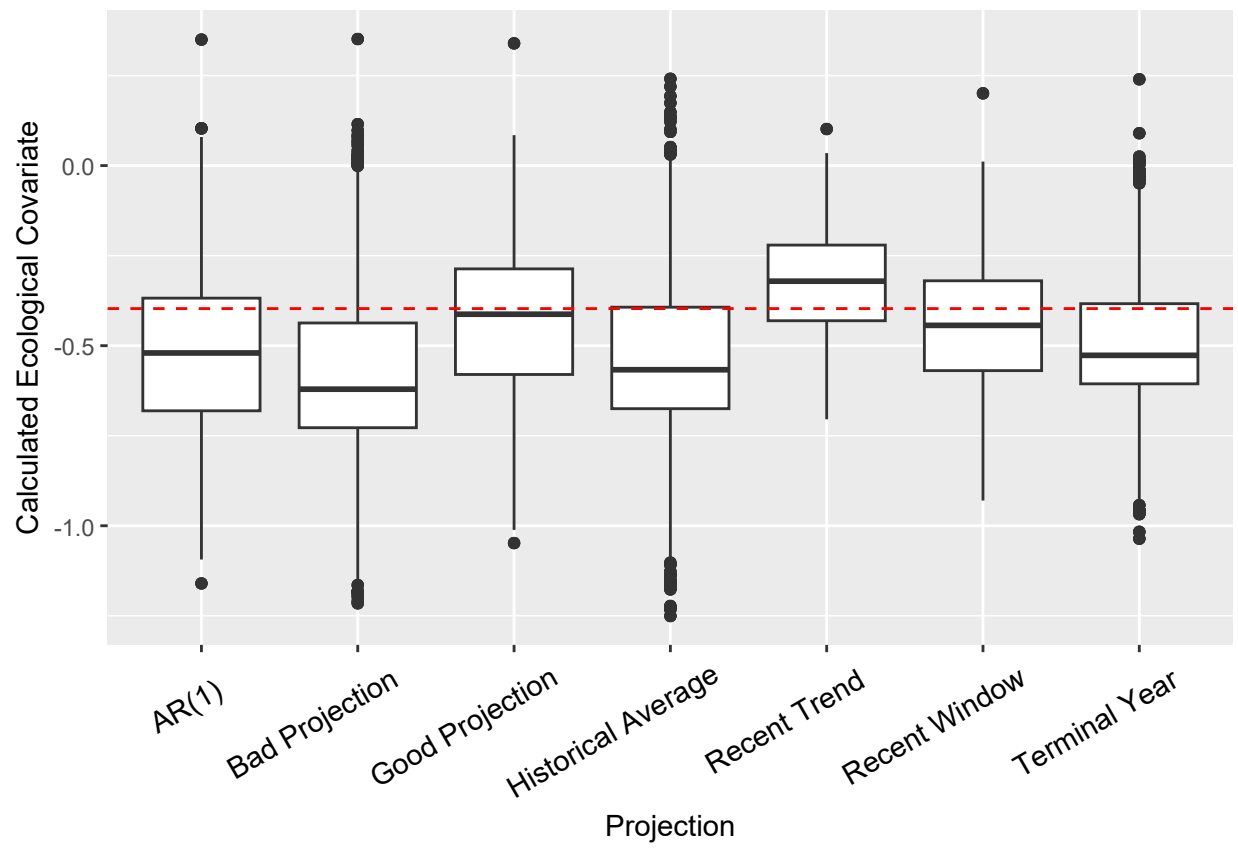


Figure 4: Estimated ecological covariates for the different biased temperature projections with a 5-year assessment cycle

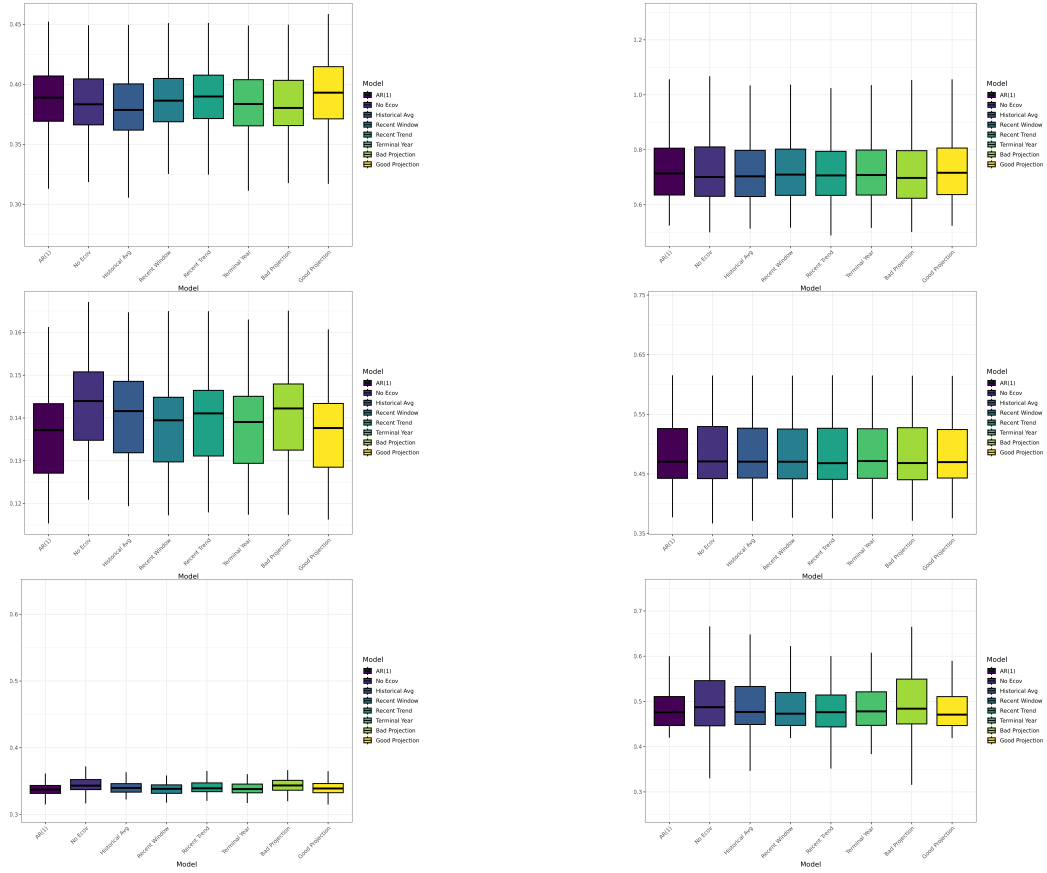


Figure 5: Catch (top), SSB (middle), and fishing mortality (bottom) for reduced variation on Numbers at Age (left) and a five-year assessment period (right)